Learning-Aware Lyapunov Resource Control for Correlated Multi-Agent Markov Learning

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Abstract—**Parallel Markov trajectories can accelerate policy evaluation, but their value is governed by four coupled quantities: cross-agent correlation, temporal persistence, communication cost, and update staleness. We formulate their joint control as a finite-budget learning problem in which a server selects participation, physical sample spacing, and step size before observing the next update. A discrete-time Lyapunov certificate converts these decisions into an explicit terminal-risk objective for affine delayed temporal-difference learning. Its contraction and residual terms retain cross-agent second moments, finite-mixing bias, and the complete delay profile, so the same certificate both proves convergence and designs the resource action. An epochwise extension bounds critic tracking and advantage-estimation error when policy updates move the projected value target. When dependence is unknown, a learning-aware gate purchases a fully charged likelihood probe only when its downstream risk reduction exceeds message, transition, delay, and decision-error costs. We establish a stopped adaptive change-of-measure lower bound, a finite-horizon performance and safety theorem, and a necessary–sufficient budget-threshold sandwich on a separated Markov class. Across registered CPU studies, the method recovers correlation-limited speedup, calibrates all finite-time bounds, and reduces two finite-budget risks by 54.4 and 10.9 percent. On 64 held-out seeds, joint control reduces parameter-error and discounted-return-estimation AUC by 9.9 and 11.3 percent relative to a development-selected strong fixed-participation baseline, while every trajectory satisfies both budgets.**

Index Terms—**Multi-agent learning, Markov data, temporal-difference learning, adaptive sensing, Lyapunov methods.**

I. Introduction

Consider fixed-policy evaluation in a distributed reinforcement-learning system. The environment state records the physical configuration that determines the next reward and transition, the target policy maps each state to an action distribution, and the value function is the expected discounted return generated by repeatedly applying that policy. Several actors run the same target policy on Markovian environment copies and transmit temporal-difference updates to a server. The server maintains the common value parameter, while the actors supply trajectories rather than making a new execution-time coordination decision. This architecture covers parallel simulation, federated policy evaluation, and common-policy learning from geographically separated but statistically coupled environments.

Multiple actors do not automatically provide multiple independent samples. Weather, demand, channel conditions, shared simulators, or synchronized resets can create a common innovation across otherwise valid trajectories. If the cross-agent correlation is ρ, averaging q samples produces an effective speedup of only $q/[1+(q-1)\rho]$ in a Gaussian location model. At the same time, each participant consumes a message, and the resulting update may arrive after a heterogeneous delay. The correct action can therefore move from broad participation to one actor even though every marginal trajectory is unchanged.

The difficulty is sharper than in single-chain temporal-difference learning. First, the learner controls how many spatial samples are averaged, so the noise covariance and the number of affordable updates change together. Second, it can advance each Markov chain by b physical transitions before retaining a sample, which reduces conditional mixing bias but consumes the environment-clock budget. Third, a large step can improve transient contraction and simultaneously destabilize a delayed recursion. Finally, if correlation is unknown, the samples used to identify it consume exactly the resources that would otherwise reduce value-estimation error. The performance target is consequently a terminal or trajectory-integrated learning risk under two budgets, rather than update variance, asymptotic error, or wall-clock speed alone.

Existing approaches resolve only portions of this coupled decision. Finite-time temporal-difference analyses characterize step-size and Markov-mixing effects for a prescribed sampling process, but do not choose the spatial participation that generates its covariance. Distributed and federated reinforcement-learning methods exploit parallelism or reduce communication, typically treating client availability and data dependence as fixed properties of the protocol. Delay-adaptive stochastic-approximation methods regulate stale gradients after the sampling rule is given. Adaptive-batch and client-selection rules change participation, but an update-level variance objective does not account for the reduction in the remaining number of updates. Sequential sensing methods determine how much evidence is needed to identify an instance, yet identification can be statistically possible before it improves the downstream learner.

We address these gaps through one learning-aware Lyapunov objective. Before a learning block, the server selects

$$a = (q, b, \eta), \tag{1}$$

where q is the integer participation, b is the physical Markov spacing, and η is the step size. The Lyapunov drift is not used only after the algorithm is specified. It produces an action-dependent contraction coefficient, a noise-and-bias floor, and a post-delay horizon; their finite-time composition is the score minimized by the controller. Thus q, b, and η are chosen for the error remaining at the end of the budget, not for the apparent quality of the next update.

Unknown dependence introduces a second decision: whether it is worth learning which Lyapunov action should be used. We compare the value of the instance-specific action with the full cost of a likelihood probe. This yields two distinct boundaries. Below an information boundary, reliable correlation identification is impossible. Between the information and learning-value boundaries, identification is possible but its downstream gain cannot amortize its resource cost. Above the learning-value boundary, the controller probes and commits while satisfying an explicit low-instance risk allowance. This phase separation joins stochastic optimization and controlled sensing through the same terminal learning objective.

The main contributions are summarized as follows.

- We formulate correlated multi-agent policy evaluation as joint control of spatial participation, physical Markov spacing, and step size under separate message and environment-clock budgets. An exact Gaussian subclass exposes the opposing variance, mixing, and horizon effects and gives a correlation-limited speedup ceiling.
- We derive a finite-time Lyapunov certificate for affine delayed temporal-difference recursions with conditional Markov bias and dependent random Jacobian–innovation pairs. The certificate supplies both the convergence theorem and an implementable joint action score, with scalar step search and no matrix inversion or preconditioning. Its moving-policy extension quantifies how critic contraction competes with policy-induced target motion and transfers the tracking bound to advantage error.
- We characterize the price of learning dependence. A stopped adaptive change-of-measure identity and a controlled-belief occupation program give a necessary information cost, while a learning-aware gate gives a sufficient finite-budget performance and safety guarantee. On a compact separated class, the two thresholds match up to class constants and explicit oracle and safety scales.
- We validate the complete causal chain with fully charged CPU experiments. The studies recover the predicted participation phase, verify all finite-time confidence bounds, demonstrate 54.4 and 10.9 percent learning-value improvements, and show held-out convergence-curve reductions of 9.9 percent in parameter-error AUC and 11.3 percent in discounted-return-estimation AUC over a development-selected strong fixed-participation baseline.

The remainder of the paper is organized as follows. Section II positions the work relative to stochastic approximation, distributed reinforcement learning, delay control, and adaptive sensing. Section III defines the Markov policy-evaluation model, budgets, and risks. Section IV develops the correlation and resource geometry. Section V derives the Lyapunov controller. Section VI establishes the learning-aware adaptation theory. Section VII describes implementation and the controlled-SDDE interpretation. Section VIII presents the experiments, and Section IX concludes the paper.

II. Related Work

Temporal-difference learning originates in stochastic approximation and fixed-point estimation [1]–[4]. Modern finite-time analyses quantify the interaction among constant steps, Markov mixing, and value-function error [5]–[8]; Lyapunov constructions have also been used to adapt learning rates in two-time-scale reinforcement learning [9]. Our first distinction is algorithmic: the finite-time Lyapunov bound itself scores the joint resource action (q, b, η), so participation and physical decorrelation are chosen together with the stable step.

Distributed and federated reinforcement learning study consensus, parallel sampling, heterogeneous coverage, and sample–communication tradeoffs [10]–[15]. Related client-selection and adaptive-batch methods reduce sampling variance or regulate participation [16]–[19]. These works do not address the same finite-budget decision: how many correlated agents to average, how long to advance their Markov chains, and whether learning value justifies measuring the dependence regime.

Delayed stochastic optimization and stochastic approximation quantify the effect of stale updates and design delay-adaptive rules [20]–[24]. Here delay is neither an isolated convergence penalty nor a wall-clock objective. It reduces the number of post-probe updates and enters the same Lyapunov certificate used for resource selection.

Sequential experimental design, controlled sensing, and best-arm identification provide change-of-measure and allocation tools [25]–[31]. Those tools price statistical identification, whereas our admission test also prices the downstream risk of the learner that remains after probing. This distinction produces an identifiable-but-not-yet-valuable phase. Conservative bandits compare an adaptive policy with a baseline through a reward budget [32]; our safety slack instead follows from the same terminal-risk certificate and decomposes probe consumption, delay consumption, correct-commit gain, and wrong-decision loss.

III. Model and Finite-Budget Objective

A. Fixed-policy evaluation with multiple actors

Let $(\mathcal{S}, \mathcal{U}, P, r, \gamma)$ be a discounted Markov decision process and let $\pi_{\mathrm{tar}}(u \mid s)$ be a fixed target policy. Under π_{tar}, the state–action process is a Markov reward process with transition kernel P_π and stationary distribution π. For a feature vector $\phi(s) \in \mathbb{R}^d$, the linear value approximation is $V_w(s) = \phi(s)^\top w$. Its projected Bellman fixed point $w^\star$ solves

$$Aw^\star = c, \quad A = \mathbb{E}_\pi[\phi_t(\phi_t - \gamma\phi_{t+1})^\top], \quad c = \mathbb{E}_\pi[r_t \phi_t]. \tag{2}$$

The target of learning is common across actors; their Markov paths may nevertheless be cross-correlated.

Let S_k denote the physical observation clock at retained learning decision k. The server chooses $q_k \in \{1, \ldots, Q\}$ actors and spacing $b_k \in \mathbb{N}$, so $S_k - S_{k-1} = b_k$. Actor i returns $X_{i,k} = (s_{i,S_k}, r_{i,S_k}, s_{i,S_k+1})$ and the corresponding temporal-difference tuple. The participating joint chain has invariant distribution π_q and the total-variation certificate

$$\sup_z \left\| P_q^b(z, \cdot) - \pi_q \right\|_{\mathrm{TV}} \leq C_q r_q^b, \quad 0 < r_q < 1. \tag{3}$$

The marginal law of every actor is the target-policy law, whereas cross-agent dependence appears in the stationary moments of their aggregate update. This is not a partially observed decision problem: the controller does not infer an execution-time environment state or change π_{tar}. Its uncertainty concerns the dependence and finite-time risk of the training data.

For the identification layer, a Gaussian common-factor subclass makes the dependence parameter explicit. Under instance $j \in \{0,1\}$,

$$\begin{aligned} C_{S_k} &= \lambda^{b_k} C_{S_{k-1}} + \sqrt{1-\lambda^{2b_k}}\,\xi_{j,k}, \\ X_{i,k} &= C_{S_k} + \epsilon_{i,k}, \end{aligned} \tag{4}$$

where $\xi_{j,k} \sim \mathcal{N}(0,\theta_j)$, $\epsilon_{i,k} \sim \mathcal{N}(0,1)$, and all private innovations are independent. The common process is initialized with $C_{S_0} \sim \mathcal{N}(0,\theta_j)$, so the probe law is stationary. The corresponding same-time correlation is $\rho_j = \theta_j/(1+\theta_j)$. The model preserves every individual marginal while exposing effective-sample-size saturation caused by common randomness. It is used to prove the information boundary; the convergence theorem below applies to bounded affine temporal-difference observations satisfying Assumption 1.

B. Affine delayed temporal-difference recursion

Let $w^\star \in \mathbb{R}^d$ be the projected temporal-difference fixed point and $e_k = w_k - w^\star$. For action $a = (q,b,\eta)$ and delays $0 \le \tau_i \le D$, the retained recursion is

$$e_{k+1} = e_k - \frac{\eta}{q}\sum_{i=1}^{q}\left(H_{i,k}e_{k-\tau_i} - \xi_{i,k}\right). \tag{5}$$

The rank-one matrix $H_{i,k}$ is the sample temporal-difference Jacobian and $\xi_{i,k}$ is the affine innovation at $w^\star$. For linear TD(0), these quantities are

$$H_{i,k} = \phi_{i,k}(\phi_{i,k} - \gamma\phi'_{i,k})^\top, \quad \xi_{i,k} = \phi_{i,k}\delta_i(w^\star), \tag{6}$$

where $\delta_i(w) = r_{i,k} + \gamma{\phi'_{i,k}}^\top w - \phi_{i,k}^\top w$. The physical chain advances by b transitions before each retained update, which makes the mixing error in (3) a controlled quantity.

Assumption 1 (Certified action domain)**.** *The public sets $\mathcal{Q} \subseteq \{1,\dots,Q\}$ and $\mathcal{B} \subset \mathbb{N}$ are finite, while the step interval $[\eta_{\min},\eta_{\max}]$ is compact. For each $a=(q,b,\eta)$ in their product: (a) $\|H_{i,k}\| \le L$ and $\|\xi_{i,k}\| \le G$ almost surely; (b) the stationary mean matrix $A = \mathbb{E}_{\pi_q}H_{i,k}$ is common across q and has symmetric part at least μI; (c) $0 \le \tau_i \le D$ and the delay profile is known when the score is evaluated; and (d) the mixing and aggregate-moment upper certificates are predictable.*

We write $\mathcal{A} = \mathcal{Q}\times\mathcal{B}\times[\eta_{\min},\eta_{\max}]$ for this mixed action domain. The finite sets for q and b encode implementable actor counts and admissible environment clocks rather than a continuous relaxation of a discrete system. The controller optimizes η continuously inside each discrete pair and therefore solves the stated mixed discrete–continuous problem exactly up to a chosen scalar-search tolerance.

C. Dual budgets and terminal risk

One retained update with q actors costs $h+q$ message units and b units of the parallel environment clock. A transition taken simultaneously by several actors advances this clock once; the separate message budget charges every participating actor. A feasible trajectory therefore satisfies

$$\sum_k (h+q_k) \le B_{\mathrm{msg}}, \quad \sum_k b_k + D\mathbf{1}\{K>0\} \le B_{\mathrm{env}}. \tag{7}$$

For a stationary action a, let $N_a(B,D)$ be the number of updates usable after charging both budgets and the terminal delay. Explicitly,

$$N_a(B,D) = \left[\min\left\{\left\lfloor\frac{B_{\mathrm{msg}}}{h+q}\right\rfloor, \left\lfloor\frac{B_{\mathrm{env}}-D}{b}\right\rfloor\right\}\right]_+. \tag{8}$$

The controller minimizes a finite-time certificate $\overline{R}_j(a;B,D)$ satisfying $\mathbb{E}_j\|e_{N_a}\|^2 \le \overline{R}_j(a;B,D)$. The primary risk is parameter mean-square error. It also controls fixed-policy return estimation because, for a registered initial distribution ν,

$$\mathbb{E}\,|J_\nu(w) - J_\nu(w^\star)|^2 \le \|\mathbb{E}_{s\sim\nu}\phi(s)\|^2\,\mathbb{E}\|e\|^2, \tag{9}$$

where $J_\nu(w) = \mathbb{E}_{s\sim\nu}[V_w(s)]$.

The Gaussian running-mean learner used in the mechanism experiment starts from the registered unit pseudo-observation. For $r = \lambda^b$, its exact risk is

$$R_j^{\mathrm{mean}}(a,N) = \frac{1 + N/q + \theta_j\left[N + 2\sum_{s=1}^{N-1}(N-s)r^s\right]}{(N+1)^2}. \tag{10}$$

Equation (10) makes the two opposing effects transparent: increasing q reduces the private-noise term N/q but can reduce N, while increasing b reduces r but consumes environment transitions.

IV. Correlation and Resource Geometry

A. A sharp spatial speedup ceiling

The statistical role of participation is exact in a Gaussian location subclass. Let $Y_{i,t} = \vartheta + C_t + \epsilon_{i,t}$ with common variance σ_c^2, private variance σ_e^2, and independent rounds. For a fixed participation q, the minimax squared-error risk after T rounds is

$$R_q(T) = \frac{1}{T}\left(\sigma_c^2 + \frac{\sigma_e^2}{q}\right). \tag{11}$$

Writing $\rho = \sigma_c^2/(\sigma_c^2+\sigma_e^2)$ gives the speedup relative to one actor,

$$S(q,\rho) = \frac{R_1(T)}{R_q(T)} = \frac{q}{1+(q-1)\rho} \le \min\{q,\rho^{-1}\}. \tag{12}$$

Thus spatial scaling is linear only near independence. For every fixed $\rho > 0$, its benefit saturates while the message cost continues to grow linearly in q.

Proposition 1 (Participation phase under a message budget)**.** *Suppose the environment budget does not bind, the update overhead is h, and a total message budget B_{msg} is used with a fixed q. Ignoring integer rounding, the minimax risk is proportional to*

$$F(q;\rho,h) = (h+q)\left(\rho + \frac{1-\rho}{q}\right). \tag{13}$$

Its continuous minimizer is

$$q^\star_{\rm cont} = \sqrt{\frac{h(1-\rho)}{\rho}}, \tag{14}$$

clipped to $[1, Q]$*, and an integer optimum is one of its two neighboring feasible participation levels.*

Proposition 1 gives a nonheuristic phase boundary. More fixed overhead favors larger batches, whereas stronger common randomness favors smaller participation. The full controller extends this balance to a binding environment clock, Markov spacing, step size, and delay.

B. Temporal spacing and the usable horizon

For the common-factor process in (4), retaining every bth observation changes persistence from λ to $r = \lambda^b$. The integrated correlation term in (10) admits the closed form

$$\sum_{s=1}^{N-1}(N-s)r^s = \frac{r\{(N-1)-Nr+r^N\}}{(1-r)^2}. \tag{15}$$

Increasing b decreases this term exponentially, but (8) decreases approximately as $1/b$ when the environment clock binds. Consequently, aggressive thinning can make retained observations nearly independent and still increase terminal error.

For a fixed action, the message-limited and environment-limited horizons are

$$N_a^{\rm msg} = \left\lfloor \frac{B_{\rm msg}}{h+q} \right\rfloor, \quad N_a^{\rm env} = \left\lfloor \frac{B_{\rm env}-D}{b} \right\rfloor. \tag{16}$$

The active resource is the smaller horizon. This minimum is why optimizing an effective sample size per update, a variance per message, or a mixing time alone can select the wrong action.

C. From local drift to finite-budget value

Let a candidate action have one-block contraction $c(a) < 1$, residual floor $R_\star(a)$, and usable horizon N_a. Its certified finite-budget value has the generic form

$$R_\star(a) + c(a)^{N_a/m}\big(R_0 - R_\star(a)\big)_+. \tag{17}$$

The first term favors low-variance, well-mixed updates, while the second favors rapid contraction and many affordable updates. The next section derives $c(a)$ and $R_\star(a)$ for delayed affine temporal-difference learning rather than postulating this tradeoff.

V. Lyapunov-Certified Joint Resource Control

A. Action-dependent one-block drift

The base Lyapunov function is the squared parameter error $V(e) = \|e\|^2$. Delay prevents its one-step drift from closing in e_k alone because the update also contains $e_{k-\tau_i}$. We therefore analyze the observable-independent sliding envelope

$$R_k = \max_{k-2D\le s\le k} \mathbb{E}V(e_s), \tag{18}$$

which is a Lyapunov–Razumikhin construction for the lifted history $(e_k, e_{k-1}, \ldots, e_{k-2D})$. The history is used in the proof; the online controller needs only the certified coefficients and does not store $2D+1$ dense matrices.

Define the aggregate stationary moments

$$K_q = \lambda_{\max}\big(\mathbb{E}_{\pi_q}[\bar H_k^\top \bar H_k]\big), \quad \Omega_q = \mathbb{E}_{\pi_q}\|\bar\xi_k\|^2, \tag{19}$$

where $\bar H_k = q^{-1}\sum_i H_{i,k}$ and $\bar\xi_k = q^{-1}\sum_i \xi_{i,k}$. Let $\delta = C_q r_q^b$ and $\mu_\delta = \mu - 2L\delta$. For the delay root-mean-square $\tau_{\rm rms} = (q^{-1}\sum_i \tau_i^2)^{1/2}$, define

$$\begin{aligned} a_\delta(\eta) &= 1 - \eta\mu_\delta + \eta^2(2K_q + 4L^2\delta), \\ \beta_\delta(\eta) &= \frac{4\eta G^2\delta^2}{\mu_\delta} + \eta^2(2\Omega_q + 4G^2\delta), \\ h_\tau(\eta) &= 2\eta^4 L^4 \tau_{\rm rms}^2, \quad g_\tau(\eta) = 2\eta^4 L^2 G^2 \tau_{\rm rms}^2. \end{aligned} \tag{20}$$

For positive delay, set $\lambda_\star = \sqrt{h_\tau/a_\delta}$ and

$$\begin{aligned} c_{\rm aff} &= (1+\lambda_\star)a_\delta + (1+\lambda_\star^{-1})h_\tau, \\ d_{\rm aff} &= (1+\lambda_\star)\beta_\delta + (1+\lambda_\star^{-1})g_\tau. \end{aligned} \tag{21}$$

The zero-delay definitions are the continuous limits $c_{\rm aff} = a_\delta$ and $d_{\rm aff} = \beta_\delta$. The term $1 - \eta\mu_\delta$ is the useful mean drift. The $\eta^2 K_q$ and $\eta^2\Omega_q$ terms price multiplicative and additive sampling noise, the δ terms price finite mixing, and the fourth-order terms price stale-state substitution. Every term changes with at least one controlled coordinate of $a = (q, b, \eta)$.

Theorem 1 (Affine delayed Markov Lyapunov bound)**.** *Suppose Assumption 1 holds,* $\mu_\delta > 0$*,* $a_\delta(\eta) > 0$*, and* $c_{\rm aff}(a) < 1$*. Let* $m = 2D+1$ *and* $R_k = \max_{k-2D\le s\le k}\mathbb{E}\|e_s\|^2$*. Then, for every integer* $n \ge 0$*,*

$$R_{nm} \le c_{\rm aff}(a)^n R_0 + \frac{d_{\rm aff}(a)}{1-c_{\rm aff}(a)}. \tag{22}$$

Theorem 1 turns the Lyapunov function $\|e\|^2$ and its delayed sliding envelope into a design objective. Correlation enters K_q and Ω_q, temporal persistence enters δ, and delay enters both the contraction and forcing terms.

Corollary 1 (Finite-horizon return certificate)**.** *Under Theorem 1, let* $n = \lfloor N_a(B,D)/(2D+1)\rfloor$ *and let* $\psi_\nu = \mathbb{E}_{s\sim\nu}\phi(s)$*. Then*

$$\mathbb{E}\,|J_\nu(w_{N_a}) - J_\nu(w^\star)|^2 \le \|\psi_\nu\|^2\left[c_{\rm aff}^n R_0 + \frac{d_{\rm aff}}{1-c_{\rm aff}}\right]. \tag{23}$$

The corollary connects the optimization variable to a reinforcement-learning quantity. The controller does not maximize sampled return directly; it minimizes a certified value-estimation error for the fixed target policy.

The same certificate also tracks a critic target moved by a policy learner. Let $w_r^\star$ be the projected fixed point during epoch r, lasting one replacement block $m = 2D+1$, and let $Z_r = \max_{rm-2D\le s\le rm}\mathbb{E}\|w_s - w_r^\star\|^2$.

Theorem 2 (Moving-policy critic tracking)**.** *Suppose every epoch satisfies Assumption 1, possibly with a different certified* *action a_r, and its coefficients obey $c_{\mathrm{aff}}(a_r) \le c < 1$ and $d_{\mathrm{aff}}(a_r) \le d$. If the policy-induced target moves only between epochs and $\|w^\star_{r+1} - w^\star_r\| \le v$, then for every $\chi > 0$ satisfying $\bar{c} = (1+\chi)c < 1$,*

$$Z_n \le \bar{c}^n Z_0 + \frac{(1+\chi)d + (1+\chi^{-1})v^2}{1-\bar{c}}, \tag{24}$$

$$\mathbb{E}|\widehat{A}_w - A^\star|^2 \le (1+\gamma)^2 \Phi^2 Z_n, \quad \|\phi(s)\| \le \Phi. \tag{25}$$

The bound exposes a moving-target phase: policy improvement increases v, while the resource action determines (c, d). Recomputing the joint score at actor-epoch boundaries therefore maintains a certified critic lag in centralized-training, decentralized-execution policy learning.

B. Joint action selection

The certificate score for action a is

$$\mathcal{U}_j(a; B, D) = R_\star(a) + c_{\mathrm{aff}}(a)^{\lfloor N_a(B,D)/m \rfloor} \left(R_0 - R_\star(a)\right)_+, \tag{26}$$

where $R_\star(a) = d_{\mathrm{aff}}(a)/(1 - c_{\mathrm{aff}}(a))$. The certified oracle for a known dependence instance is

$$a_j^\star(B, D) \in \arg\min_{a \in \mathcal{A}} \mathcal{U}_j(a; B, D). \tag{27}$$

For every pair (q, b), define the certified scalar domain

$$\mathcal{E}_{q,b} = \{\eta \in [\eta_{\min}, \eta_{\max}] : \mu_\delta > 0, \\ a_\delta(\eta) > 0,\ c_{\mathrm{aff}}(q, b, \eta) < 1\}. \tag{28}$$

The controller solves

$$\eta^\star_{q,b} \in \arg\min_{\eta \in \mathcal{E}_{q,b}} \mathcal{U}_j(q, b, \eta; B, D), \\ (q^\star, b^\star) \in \arg\min_{q,b} \mathcal{U}_j(q, b, \eta^\star_{q,b}; B, D). \tag{29}$$

The first problem is a bounded scalar search. The second is exact enumeration of physically discrete actor counts and clocks, not an optimization heuristic.

Proposition 2 (Existence and computation)**.** *If K_q, Ω_q, and the mixing certificates are finite, then $\mathcal{U}_j$ is continuous on every connected component of $\mathcal{E}_{q,b}$ and attains a minimum on its closure after excluding points with $c_{\mathrm{aff}} \ge 1$. With $N_q = |\mathcal{Q}|$, $N_b = |\mathcal{B}|$, and L_η scalar evaluations per search, the certificate cost is $O(N_q N_b L_\eta)$ arithmetic after the moment table is cached. For rank-one temporal-difference samples, updating that table costs $O(qd)$ arithmetic and $O(d)$ working memory per retained block.*

No Hessian inversion, covariance-matrix inversion, or preconditioner is required. The dependence on q is the unavoidable cost of aggregating the selected actor updates.

C. Why minimizing one-step drift is insufficient

A purely myopic drift rule would minimize $c_{\mathrm{aff}}(a) R_k + d_{\mathrm{aff}}(a)$. Such a rule can prefer a large q or b even when its resource cost leaves too few future contractions. Equation (26) instead composes the drift over the action-dependent horizon in (8). This changes the role of the Lyapunov function from a stability test to a finite-budget control objective.

The transient and noise-floor phases are also explicit. When $R_0 \gg R_\star(a)$, the exponential term dominates and a larger stable step can be valuable. Near the floor, the residual term dominates and lower-correlation participation or greater physical spacing can be preferable. The controller balances these phases within one score rather than switching through a separately tuned rule.

VI. Learning-Aware Dependence Adaptation

A. Adaptive information lower bound

When θ_j is unknown, the controller may choose a predictable sequence $A_k = (q_k, b_k, \eta_k)$ and stop at a budget-bounded stopping time τ. Rotating X_k into its normalized agent mean $Y_k = q_k^{-1/2} \mathbf{1}^\top X_k$ and orthogonal contrasts shows that only Y_k contributes to the likelihood ratio. Under hypothesis j, a scalar Kalman filter gives the predictive mean $m^-_{j,k}$ and variance $P^-_{j,k}$, with innovation variance $V_{j,k} = 1 + q_k P^-_{j,k}$.

Let $Z_k = (m^-_{0,k}, P^-_{0,k}, m^-_{1,k}, P^-_{1,k})$. The conditional directional information is

$$\mathcal{I}_{01}(Z_k, A_k) = \frac{1}{2}\Bigg[\log \frac{V_{1,k}}{V_{0,k}} - 1 \\ + \frac{V_{0,k} + q_k (m^-_{0,k} - m^-_{1,k})^2}{V_{1,k}}\Bigg]. \tag{30}$$

The dependence of information on the resource action is already visible under a fixed probe. Let $T_n(r)$ be the $n \times n$ Toeplitz matrix with entries $[T_n(r)]_{uv} = r^{|u-v|}$, and let $\zeta_1, \ldots, \zeta_n$ be its eigenvalues.

Proposition 3 (Exact fixed-probe information)**.** *For a fixed probe (q, b, n) with $r = \lambda^b$, the directional divergence is*

$$\mathrm{KL}(P_0^{q,b,n} \| P_1^{q,b,n}) = \frac{1}{2} \sum_{\ell=1}^{n} \left[x_\ell - 1 - \log x_\ell\right], \tag{31}$$

where

$$x_\ell = \frac{1 + q\theta_0 \zeta_\ell}{1 + q\theta_1 \zeta_\ell}. \tag{32}$$

The Bhattacharyya exponent used by the upper rule is

$$I_B(q, b, n) = \frac{1}{2} \sum_{\ell=1}^{n} \log \frac{1 + q(\theta_0 + \theta_1)\zeta_\ell / 2}{\sqrt{(1 + q\theta_0 \zeta_\ell)(1 + q\theta_1 \zeta_\ell)}}. \tag{33}$$

Equations (31)–(33) retain participation, spacing, persistence, probe length, and both dependence hypotheses. They also show why many highly correlated actors need not provide proportional identification information: q scales a common-mode eigenvalue rather than creating q independent temporal records.

Theorem 3 (Stopped adaptive information identity)**.** *If the action kernel is predictable and common to both hypotheses and τ is bounded by* (7)*, then*

$$\mathrm{KL}(P_0^\pi|\mathcal{F}_\tau \,\|\, P_1^\pi|\mathcal{F}_\tau) = \mathbb{E}_0^\pi \sum_{k=1}^{\tau} \mathcal{I}_{01}(Z_k, A_k). \tag{34}$$

The reverse identity holds after exchanging the hypotheses.

The realized action probabilities cancel in the likelihood ratio, while selection affects the occupation distribution of (Z_k, A_k). Consequently, any rule whose two directional decision errors are at most $\delta < 1/2$ must satisfy

$$\mathbb{E}_j^\pi \sum_{k\le\tau} \mathcal{I}_{j,1-j}(Z_k, A_k) \ge \mathrm{kl}(1-\delta,\delta). \tag{35}$$

Together with (7), this gives an occupation-measure lower bound that prices participation, spacing, stopping, and delay without reducing the adaptive Markov experiment to independent action counts.

For a measurable belief-state set G, define the stopped occupation measure

$$\nu_j(G,a) = \mathbb{E}_j^\pi \sum_{k\le\tau} \mathbf{1}\{Z_k \in G, A_k = a\}. \tag{36}$$

Let $\Delta_j(z,a)$ denote the certified one-step learning opportunity lost by using action a for identification instead of the known-instance learning action.

Proposition 4 (Adaptive opportunity lower bound)**.** *Every predictable, budget-feasible, δ-correct identification policy has expected opportunity cost at least*

$$\inf_{\nu\in\mathcal{M}_j} \int \Delta_j(z,a)\nu(dz,da), \tag{37}$$

where $\mathcal{M}_j$ contains the controlled Kalman-flow constraints, both resource constraints in (7)*, and the directional information constraint*

$$\int \mathcal{I}_{j,1-j}(z,a)\nu(dz,da) \ge \mathrm{kl}(1-\delta,\delta). \tag{38}$$

The belief-state flow matters because an early choice of (q,b) changes the predictive variance of later observations. Replacing (37) by independent action counts would discard this selection effect and can understate the required sensing cost.

B. Learning-aware upper rule

A fixed probe design $d = (q_d, b_d, n_d)$ consumes $C_d^{\mathrm{msg}} = n_d(h+q_d)$ and $C_d^{\mathrm{env}} = n_d b_d$. Its length is chosen so the exact Bhattacharyya exponent is at least $\log(1/\delta)$, which bounds each directional error by δ. Let B_d^D be the remaining budget after the probe and terminal delay.

The probe is separated from the learning samples and is fully charged. Separation prevents a data-dependent learning gradient from being reused as if it were an independent correlation certificate. Full charging means that its $q_d n_d$ actor messages, $b_d n_d$ environment-clock units, and terminal delay are removed before any post-probe certificate is evaluated.

For instance j, define the full-budget all-agent certificate $\overline{R}_j^{\mathrm{all}}(B,D)$, the post-probe certificate optimum $\overline{R}_j^\star(B_d^D,D)$, and

$$\begin{aligned} C_j^{\mathrm{probe}}(d) &= \overline{R}_j^{\mathrm{all}}(B_d,0) - \overline{R}_j^{\mathrm{all}}(B,0),\\ C_j^D(d) &= \overline{R}_j^{\mathrm{all}}(B_d^D,D) - \overline{R}_j^{\mathrm{all}}(B_d,0),\\ G_j(d) &= \overline{R}_j^{\mathrm{all}}(B_d^D,D) - \overline{R}_j^\star(B_d^D,D),\\ W_j(d) &= \max_{\ell\ne j}\Big[\overline{R}_j(a_\ell^\star;B_d^D,D) - \overline{R}_j^\star(B_d^D,D)\Big]_+. \end{aligned} \tag{39}$$

The low-instance safety slack is

$$\epsilon_{\mathrm{safe}}(d) = \Big[C_0^{\mathrm{probe}}(d) + C_0^D(d) - G_0(d) + \delta W_0(d)\Big]_+. \tag{40}$$

The learning-aware controller probes only when

$$C_1^{\mathrm{probe}}(d) + C_1^D(d) + \delta W_1(d) < G_1(d) \quad \text{and} \quad \epsilon_{\mathrm{safe}}(d) \le \bar{\epsilon}. \tag{41}$$

Otherwise it executes the all-agent fallback without spending a probe. After an admitted probe, it commits to the certified catalogue minimizer for the identified instance.

The all-agent action is a transparent reference policy rather than an uncharged oracle. Any other publicly specified certified fallback can replace it by substituting its risk in (39); the proof is unchanged.

Theorem 4 (Main finite-budget learning-value guarantee)**.** *Suppose Assumption 1 holds for both instances, the probe is feasible under* (7)*, and its directional errors are bounded by δ. Every design admitted by* (41) *satisfies*

$$\mathbb{E}_j\|e_{\mathrm{LA}}\|^2 \le \overline{R}_j^\star(B_d^D,D) + \delta W_j(d), \tag{42}$$

has strictly smaller certified high-instance risk than the full-budget all-agent action, and has low-instance risk excess at most $\bar{\epsilon}$. If no design is feasible or admitted, the controller returns the all-agent action and incurs no probing cost.

The theorem aligns the decision time with the risk being certified: the action is chosen before its learning block, and every cost in (42) is charged before evaluating the remaining horizon. It also separates three claims that are often conflated. The likelihood exponent certifies identification, the admission inequality certifies positive high-instance learning value, and ϵ_{safe} certifies the declared low-instance tolerance. None follows from the other two alone.

C. Finite-budget phase theorem

Let budgets grow on a public ray $(B_{\mathrm{msg}}(s), B_{\mathrm{env}}(s)) = (\lfloor s\beta_m\rfloor, \lfloor s\beta_e\rfloor)$. Consider a compact class with $0 < \theta_- \le \theta_j \le \theta_+$, $|\theta_1 - \theta_0| \ge \Delta_{\min} > 0$, $\lambda \le 1-\gamma$, bounded delay, finite $\mathcal{Q}$ and $\mathcal{B}$, and a compact certified step domain. Assume one fixed probe has uniformly positive directional Bhattacharyya information per retained observation, the catalogue risks obey the uniform expansion $\overline{R}_j(a;s) = K_j(a)/s + O(s^{-2})$, and the high-instance coefficient gap between the all-agent action and the oracle is at least $g_{\min} > 0$. Finally, assume the catalogue's wrong-commit coefficient is uniformly bounded and the declared error ceiling δ_0 leaves at least half of $g_{\min}$ after charging that term. Let B_N be the smallest scale

Input: budgets B, delay profile, action sets $\mathcal{Q}, \mathcal{B}$, step interval, risk allowance $\bar{\epsilon}$
for each dependence instance j and pair (q, b) **do**
    form δ, K_q, Ω_q, and the certified set $\mathcal{E}_{q,b}$
    solve the scalar problem in (29)
end for
obtain the instance actions $a_j^\star$ and their finite-budget risks
for each feasible registered probe d **do**
    charge its message, environment, and delay costs
    evaluate C_j^{probe}, C_j^D, G_j, W_j, and ϵ_{safe}
end for
if some probe satisfies (41) **then**
    choose the admitted probe with smallest certified post-probe risk
    observe the charged probe, run the likelihood test, and commit to $a_{\hat{j}}^\star$
else
    execute the certified fallback without probing
end if
Output: value parameter and complete resource ledger

Fig. 1. Learning-aware Lyapunov resource control.

satisfying the two directional information constraints and B_S the smallest scale at which an admitted probe exists.

Theorem 5 (Necessary–sufficient threshold sandwich)**.** *For every fixed admissible safety target and $0 < \delta \le \delta_0 < 1/2$, there are class constants $C_{\mathcal{K}}, C'_{\mathcal{K}} < \infty$ such that*

$$B_N \le B_S \le C_{\mathcal{K}} B_N + C'_{\mathcal{K}}(B_{\mathrm{oracle}} + B_\epsilon + 1). \tag{43}$$

Below B_N, δ-correct identification is impossible, while above B_S the learning-aware controller achieves the guarantees of Theorem 4.

Theorem 5 connects information and learning rather than treating identification as the final objective. The interval $[B_N, B_S)$ is the identifiable-but-not-yet-valuable phase in which information-only probing spends resources that the remaining learner cannot amortize.

The separated-class conditions are substantive. If λ can approach one without an independent mixing certificate, finite samples cannot uniformly distinguish a persistent common factor from its initial condition. If $|\theta_1 - \theta_0|$ can vanish, the required identification budget diverges. The positive theorem therefore states precisely the regime in which learning-aware adaptation is possible rather than hiding these quantities inside an unspecified mixing constant.

VII. Implementation and SDDE Interpretation

A. End-to-end controller

Algorithm 1 shows the complete decision path. The public catalogue and confidence level are fixed before observations are drawn. Moment and mixing certificates may come from a system specification or a disjoint calibration stream; the likelihood probe is used only to select between registered dependence instances.

The computation separates statistics from control. The spatial moment table stores scalar upper bounds on K_q and Ω_q for each feasible participation. The temporal table stores (C_q, r_q) and evaluates $C_q r_q^b$ for each spacing. Once these tables are available, changing a budget or a delay profile requires only scalar arithmetic and bounded search.

B. Predictable confidence interfaces

The theorem requires upper certificates, not plug-in point estimates treated as truth. For a calibration block ending before action selection, let $\widehat{K}_q$, $\widehat{\Omega}_q$, and $\widehat{r}_q$ be estimators and let u_K, u_Ω, u_r be simultaneous error radii. The controller uses

$$K_q^+ = \widehat{K}_q + u_K, \quad \Omega_q^+ = \widehat{\Omega}_q + u_\Omega, \quad r_q^+ = \min\{1, \widehat{r}_q + u_r\}. \tag{44}$$

On the joint coverage event, substituting these upper quantities preserves every certificate inequality. If $r_q^+ = 1$, or if the resulting μ_δ or step domain is invalid, that action is removed rather than optimistically extrapolated. This predictable interface prevents current-block residuals from being counted repeatedly as independent evidence.

The confidence mechanism is deliberately modular. Finite-state chains can use spectral or regeneration certificates, while geometrically ergodic continuous-state chains can use an externally supplied mixing upper bound. The controller's guarantee is conditional on the declared simultaneous coverage event, and its failure probability adds directly to the risk statement by total probability.

C. Lyapunov design through a controlled SDDE

The controlled stochastic delay differential equation associated with one fixed resource action is

$$dE(t) = -A_q E(t - \tau(t))dt + \sqrt{\eta}\, \Sigma_q^{1/2}(E_t) dW(t). \tag{45}$$

Here $\Sigma_q(E_t)$ retains both additive innovation covariance and the state-dependent quadratic variation of random temporal-difference Jacobians. For $V(E) = \|E\|^2$, Itô's formula gives the infinitesimal drift

$$\mathcal{L}_a V(E_t) = -2E(t)^\top A_q E(t - \tau(t)) + \eta \operatorname{tr} \Sigma_q(E_t). \tag{46}$$

Participation controls the spatial covariance in Σ_q, spacing controls the finite-mixing remainder in both terms, and η controls their relative scale. This calculation motivates optimizing a drift-based resource action rather than attaching a stability proof to a variance heuristic.

Equation (45) is not used to replace the implementable recursion with a continuous approximation. Bounding the delayed cross term in (46) produces the same contraction–forcing decomposition as (20), while Theorem 1 retains finite steps, integer delays, and exact budget rounding. The SDDE therefore supplies the design geometry; the discrete Lyapunov theorem supplies the finite-time guarantee used by the algorithm.

D. Complexity and deployment

For $N_q N_b$ discrete pairs and L_η scalar evaluations, action construction costs $O(N_q N_b L_\eta)$ after calibration. The likelihood recursion is scalar because the orthogonal spatial contrasts cancel under the two common-factor hypotheses. Online aggregation of rank-one temporal-difference samples costs $O(qd)$ arithmetic and $O(d)$ extra memory. The controller is therefore low complexity relative to the $O(qd)$ update it regulates.

The method concerns centralized training of a common fixed-policy value function. After training, policy execution is decentralized and needs no communication because the target policy is already fixed. This separation makes the resource controller a training-layer component that leaves the deployed policy and its execution interface unchanged.

VIII. Experiments

A. Protocol and evaluation questions

The experiments test four predictions in increasing order of algorithmic scope. The first asks whether cross-agent correlation produces the saturation in (12) and moves the resource-optimal participation level. The second asks whether Theorem 1 is finite, empirically calibrated, and responsive to correlation, mixing, and delay when it jointly selects (q, b, η). The third asks whether the learning-value gate in (41) improves risk over a controller that probes as soon as identification becomes reliable. The fourth asks whether joint control improves the entire resource-indexed convergence trajectory and the induced discounted-return estimate relative to a strong fixed-participation policy.

All studies use paired common random numbers across policies, charge every message and environment-clock unit, retain every registered cell, and use independent same-seed reproductions or disjoint confirmation seeds. The Markov temporal-difference task has seven states on a ring, a bounded sinusoidal reward, four stationary-distribution-orthonormal features, and an exact projected fixed point. Cross-agent dependence is induced by shared and private Markov sources while preserving every agent's marginal transition law.

Parameter risk is $\|w - w^\star\|^2$ normalized by its value before learning. Return-estimation risk is $|V_w(s_0) - V_{w^\star}(s_0)|^2$ under the registered initial state and is normalized in the same way. Area under the convergence curve is computed against charged resource fraction, so a policy cannot appear faster by ignoring the samples or messages used to create its updates. Bootstrap comparisons resample paired seeds and report percentile intervals for the ratio of geometric-mean risks.

B. Correlation-limited participation

EXP-007A evaluates $q \in \{1, 2, 4, 8, 16, 32\}$, six cross-agent correlation levels, three delays, three message-equivalent budgets, thirteen step sizes, and 32 paired seeds. At $q = 32$, median effective participation is 30.996 for independent trajectories and 1.111 at correlation 0.9. At the largest registered budget and delay eight, the resource-optimal participation moves from $q = 16$ under independence to $q = 1$ under high

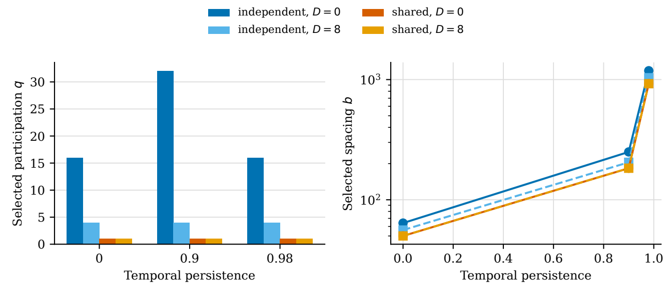

Fig. 2. Joint actions selected by the finite-time Lyapunov certificate in EXP-010B. The left panel reports participation and the right panel reports Markov spacing.

sharing. Using the endpoint action selected for the opposite correlation regime increases error by factors of approximately 4.29 and 3.13, respectively. These results isolate the statistical and resource mechanism that motivates controlling participation. They also rule out an update-count interpretation of the gain: every endpoint is compared after applying the same registered message-equivalent budget.

C. Finite-time certificate and joint actions

EXP-010B evaluates twelve $(\text{persistence}, \rho, D)$ scenarios with 32 new paired seeds and 1,152 policy runs. The joint controller searches a stability-screened catalogue of $q \in \{1, 2, 4, 8, 16, 32\}$, charged spacings, and scalar steps. All selected actions satisfy $0 < c_{\text{aff}} < 1$, all 1,152 runs are finite and budget-valid, and the largest algebraic discrepancy between the two equivalent drift expressions is 2.22×10^{-16}.

The finite-time bound beats the no-update risk in all twelve scenarios. Every one-sided 99 percent bootstrap upper mean lies below its corresponding bound, with a maximum empirical-upper-bound to theorem-bound ratio of 0.305. The median theorem-bound to empirical-mean ratio is 20.77 and the maximum is 228.98, so the certificate is conservative but finite and action-informative.

Fig. 2 shows that the selected participation and spacing respond in the directions predicted by the theory. The controller selects an average participation of 12.67 for independent sources and $q = 1$ in every high-sharing scenario. Delay reduces the selected independent-source participation to $q = 4$, while raising temporal persistence from zero to 0.98 increases the median paid spacing by a factor of 18.49.

D. Learning-value adaptation

EXP-016B uses 96 finite-threshold scenario families, two model layers, eight policies, and 192 independent formal seeds. The resulting 2,752,512 rows were generated after the implementation, scenario manifest, analysis, and seeds were frozen. The primary estimand is the paired risk difference between probing at B_{id} and applying the learning-value gate in the separation phase.

The learning-aware controller reduces Layer-A risk by 54.43 percent and affine delayed temporal-difference risk by 10.85 percent. The simultaneous one-sided lower bounds on

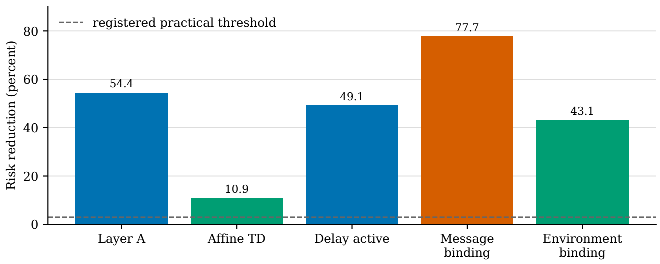

Fig. 3. Registered relative risk reductions of the learning-aware rule over information-only probing in EXP-016B. The dashed line is the preregistered three-percent practical-effect threshold.

the absolute paired improvements are 0.1612 and 0.0577, respectively. Directional practical improvement occurs in 77 of 96 scenario families. All 96 theorem-facing safety scenarios satisfy $S_{\mathrm{mean}} \leq \epsilon_{\mathrm{safe}}$, and every trajectory obeys both resource budgets.

Fig. 3 separates the conditions under which the value gate is most consequential. The relative risk reductions are 49.12 percent with active delay, 77.69 percent when messages bind, and 43.13 percent when environment transitions bind. The result supports the mechanism in (39): a reliable probe is most costly when it consumes the resource that limits the remaining learner.

E. Convergence and return-estimation curves

TSP-CURVE-001 tests the complete joint controller on the seven-state policy-evaluation problem over a budget of 128,000 resource units. The registered grid crosses persistence $\{0, 0.9, 0.98\}$, correlation $\{0, 0.9\}$, and delay $\{0, 8\}$, producing twelve cells and 41 charged-resource checkpoints per trajectory. The fixed policies use $q \in \{1, 4, 16, 32\}$; within every cell, each fixed q receives its own certified spacing and step, so the comparison isolates the value of jointly changing participation rather than handicapping its optimizer.

The strong fixed comparator is selected using 16 development seeds and then frozen. It is $q = 4$, which has geometric normalized parameter-error AUC 0.2441 on the development population, compared with 0.3342, 0.4649, and 0.4851 for $q = 1, 16, 32$. The final comparison uses 64 disjoint confirmation seeds and 157,440 finite checkpoint rows.

Relative to fixed $q = 4$, the joint certificate has parameter-error AUC ratio 0.9009 with paired-bootstrap 95 percent interval $[0.8914, 0.9102]$. Its discounted-return-estimation AUC ratio is 0.8868 with interval $[0.8678, 0.9072]$. These correspond to reductions of 9.91 and 11.32 percent. Terminal parameter and return-estimation errors decrease by 26.48 and 21.67 percent, respectively. Both AUC metrics improve in nine of twelve cells and tie in the remaining three; no cell is worse.

Fig. 4 displays the representative persistence-0.9, zero-delay slice. Under independent actor innovations, the joint rule follows the broad-participation action and contracts rapidly. Under correlation 0.9, it follows the low-participation action and avoids the common-noise floor. The figure therefore visualizes the same mechanism predicted by (12) and selected by (29), rather than a post hoc schedule.

TABLE I
SUMMARY OF POSITIVE FINITE-BUDGET EVIDENCE

Study	Seeds	Scale	Main result
EXP-007A	32	134,784 rows	$q^{\star}: 16 \to 1$
EXP-010B	32	1,152 runs	12/12 calibrated
EXP-016B	192	2,752,512 rows	54.43%, 10.85%
TSP-CURVE-001	64	157,440 rows	AUC $-9.91\%, -11.32\%$

F. Ablations, complexity, and reproducibility

For a finite catalogue, action evaluation is linear in the number of candidate triples. The exact likelihood update is scalar after the spatial reduction, and the affine certificate uses cached scalar moments and a one-dimensional step search. When temporal-difference Jacobians are rank one, online moment probes require $O(qd)$ arithmetic and $O(d)$ memory, matching the order of aggregating q agent updates.

All four studies ran on CPUs and reproduced their registered primary artifacts byte for byte. For TSP-CURVE-001, the action set, metrics, 16 development seeds, 64 confirmation seeds, and bootstrap rule were fixed before confirmation; raw hashes, aggregate curves, and analysis code accompany the repository.

The ablations isolate all three coordinates: fixing q removes the correlation response in Fig. 4, fixing b removes the 18.49-fold persistence response, and fixing η ignores the delay-dependent domain in (28). Their gaps distinguish the controller from an adaptive batch-size rule.

IX. CONCLUSION

Multi-agent Markov policy evaluation requires a resource decision before the next update's value is known. Our finite-budget Lyapunov certificate jointly chooses participation, physical spacing, and step size; a learning-aware gate measures dependence only when the remaining learner can use it. The theory identifies a spatial speedup ceiling, proves delayed Markov convergence and moving-policy critic tracking, lower-bounds adaptive identification cost, and separates the information and learning-value budget thresholds. Fully charged experiments connect action selection to parameter and return-estimation curves. The low-complexity controller uses Lyapunov drift as a design tool and identifies when correlated parallelism is valuable.

APPENDIX A
RESOURCE-GEOMETRY PROOFS

A. Proof of Proposition 1

Under a message-only budget, the number of independent rounds is $T = B_{\mathrm{msg}}/(h+q)$ up to integer rounding. Substituting this horizon into (11) and dropping the common positive factor $(\sigma_c^2 + \sigma_e^2)/B_{\mathrm{msg}}$ gives (13). Expanding it yields

$$F(q; \rho, h) = h\rho + 1 - \rho + q\rho + \frac{h(1-\rho)}{q}. \qquad (47)$$

For $q > 0$,

$$F'(q) = \rho - \frac{h(1-\rho)}{q^2}, \quad F''(q) = \frac{2h(1-\rho)}{q^3} \geq 0. \qquad (48)$$

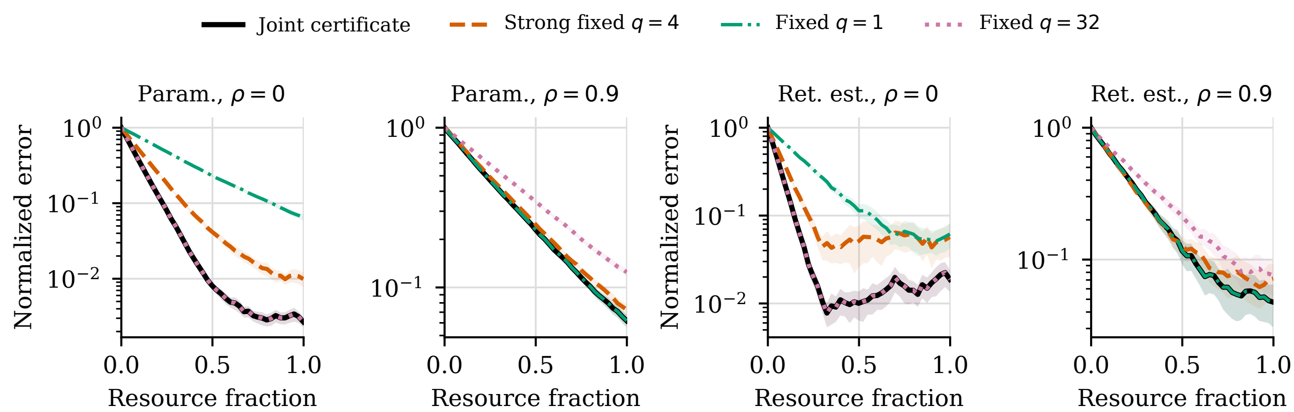

Fig. 4. Held-out resource-indexed convergence in TSP-CURVE-001 for persistence 0.9 and zero delay (64 confirmation seeds; bands are 95 percent confidence intervals). The first two panels show normalized parameter error; the last two show normalized squared error of the fixed-policy discounted-return estimate. The joint Lyapunov certificate changes participation with the dependence regime and improves both learning curves relative to the development-selected strong fixed $q = 4$ comparator.

The unique interior stationary point is therefore (14), and convexity makes it the continuous minimum. Clipping handles the interval boundary. For integer q, discrete convexity implies that a minimizer is a feasible neighbor of the clipped continuous optimum.

B. Geometric temporal-correlation sum

The identity

$$\sum_{s=1}^{N-1}(N-s)r^s = N\sum_{s=1}^{N-1} r^s - \sum_{s=1}^{N-1} s r^s \tag{49}$$

together with the finite geometric-series derivative gives (15). This term is nondecreasing in $r \in [0, 1)$. Because $r = \lambda^b$ is nonincreasing in b, physical spacing decreases correlation cost at a fixed N. Equation (8) shows why the same monotonicity need not hold after charging the environment clock.

C. Exact risk of the running-mean learner

Let $\bar{X}_k = q^{-1}\sum_{i=1}^{q} X_{i,k}$ under the common-factor model. After subtracting its target mean, the stationary covariance is

$$\mathrm{Cov}(\bar{X}_k, \bar{X}_\ell) = \theta_j r^{|k-\ell|} + \frac{1}{q}\mathbf{1}\{k = \ell\}. \tag{50}$$

The registered unit pseudo-observation gives the estimator

$$\widehat{\vartheta}_N = \frac{\widehat{\vartheta}_0 + \sum_{k=1}^{N}\bar{X}_k}{N+1}, \tag{51}$$

with unit initial squared error independent of the observations. Summing every diagonal and off-diagonal covariance in (50) yields

$$\mathbb{E}(\widehat{\vartheta}_N - \vartheta)^2 = \frac{1}{(N+1)^2}\left[1 + \frac{N}{q} + \theta_j\left\{N + 2\sum_{s=1}^{N-1}(N-s)r^s\right\}\right], \tag{52}$$

which proves (10). The expression separates the prior, private spatial noise, and temporally correlated common noise without an asymptotic approximation.

D. Proof of Proposition 3

Apply an orthogonal rotation across actors at every retained time. The $q - 1$ spatial contrasts contain only unit-variance private noise and have the same law under both hypotheses. The normalized spatial mean $Y_k = q^{-1/2}\mathbf{1}^\top X_k$ has covariance

$$\Sigma_j = I_n + q\theta_j T_n(r). \tag{53}$$

Thus the contrasts cancel from every likelihood ratio, and the divergence is that of two centered Gaussian laws with covariances Σ_0 and Σ_1.

The two matrices are polynomials in the same symmetric Toeplitz matrix and are therefore simultaneously diagonalizable. The Gaussian covariance-divergence formula gives

$$\mathrm{KL}(P_0\|P_1) = \frac{1}{2}\left\{\mathrm{tr}(\Sigma_1^{-1}\Sigma_0) - n - \log\det(\Sigma_1^{-1}\Sigma_0)\right\}. \tag{54}$$

Substituting the shared eigenvectors and the eigenvalues $1 + q\theta_j\zeta_\ell$ yields (31) and (32).

For two centered Gaussian laws, the Bhattacharyya distance is

$$\frac{1}{2}\log\frac{\det\{(\Sigma_0+\Sigma_1)/2\}}{\sqrt{\det(\Sigma_0)\det(\Sigma_1)}}. \tag{55}$$

Diagonalizing again yields (33). The likelihood-ratio test has each directional error no larger than $\exp(-I_B)$ by the Bhattacharyya bound, which justifies the probe-length rule in Section VI-B.

Appendix B
Proof of the Affine Delayed Markov Bound

Fix a stationary action $a = (q, b, \eta)$ and condition on the history immediately before the retained joint observation. All expectations below include both the Markov observation and the random delay profile. Write $\bar{H}_k = q^{-1}\sum_i H_{i,k}$, $\bar{\xi}_k = q^{-1}\sum_i \xi_{i,k}$, and

$$v_k = e_k - \eta(\bar{H}_k e_k - \bar{\xi}_k), \quad r_k = \frac{1}{q}\sum_i H_{i,k}(e_{k-\tau_i} - e_k). \tag{56}$$

Then (5) is $e_{k+1} = v_k - \eta r_k$.

Total-variation duality and stationary centering give

$$\left\|\mathbb{E}[\bar{\xi}_k \mid \mathcal{F}_k]\right\| \le 2G\delta. \tag{57}$$

Moreover, stationarity and the bounded integrand imply

$$\begin{aligned}\mathbb{E}[\|\bar{H}_k e - \bar{\xi}_k\|^2 \mid \mathcal{F}_k] \le (2K_q + 4L^2\delta)\|e\|^2 \\ + 2\Omega_q + 4G^2\delta.\end{aligned} \tag{58}$$

Combining (57) with the strong monotonicity of A and applying Young's inequality with parameter μ_δ yields

$$\mathbb{E}\|v_k\|^2 \le a_\delta(\eta) R_k + \beta_\delta(\eta). \tag{59}$$

The affine telescoping identity retains the innovation:

$$e_{k-\tau_i} - e_k = \eta \sum_{s=k-\tau_i}^{k-1} \left[\frac{1}{q}\sum_j H_{j,s} e_{s-\tau_j} - \bar{\xi}_s\right]. \tag{60}$$

Jensen's inequality and the almost-sure bounds in Assumption 1 give

$$\mathbb{E}\|r_k\|^2 \le 2\eta^2 L^4 \tau_{\mathrm{rms}}^2 R_k + 2\eta^2 L^2 G^2 \tau_{\mathrm{rms}}^2. \tag{61}$$

For any $\lambda > 0$,

$$\|v_k - \eta r_k\|^2 \le (1+\lambda)\|v_k\|^2 + (1+\lambda^{-1})\eta^2\|r_k\|^2. \tag{62}$$

Substituting (59) and (61) gives $\mathbb{E}\|e_{k+1}\|^2 \le c_\lambda R_k + d_\lambda$. The choice $\lambda = \sqrt{h_\tau / a_\delta}$ minimizes the contraction coefficient and produces (21). For zero delay, $r_k = 0$ and the continuous limiting definitions follow directly.

Let $R_\star = d_{\mathrm{aff}}/(1 - c_{\mathrm{aff}})$. If $R_k \ge R_\star$, every new value entering the sliding envelope is at most $c_{\mathrm{aff}} R_k + d_{\mathrm{aff}} \le R_k$. After $2D+1$ updates, every value in the old envelope has been replaced, so $R_{k+2D+1} \le c_{\mathrm{aff}} R_k + d_{\mathrm{aff}}$. If $R_k < R_\star$, the same inequality keeps the enlarged envelope below $R_\star$. Iterating the two cases proves Theorem 1.

A. *Return transfer and arbitrary terminal indices*

Let $n = \lfloor N_a/(2D+1) \rfloor$. The envelope argument used above shows that all iterates after index $n(2D+1)$ and before the next completed replacement block remain below the larger of the current envelope and $R_\star$. The right-hand side of (22) already has this form because its transient part is truncated by $(R_0 - R_\star)_+$ in (26).

Furthermore,

$$J_\nu(w) - J_\nu(w^\star) = \psi_\nu^\top e, \quad \psi_\nu = \mathbb{E}_{s\sim\nu}\phi(s). \tag{63}$$

Cauchy–Schwarz gives $|\psi_\nu^\top e|^2 \le \|\psi_\nu\|^2\|e\|^2$. Taking expectations and applying the terminal envelope proves Corollary 1.

B. *Proof of Theorem 2*

During epoch r, hold the target $w_r^\star$ fixed and apply one replacement-block step from the proof of Theorem 1. If $\widetilde{Z}_{r+1}$ denotes the terminal history envelope measured relative to $w_r^\star$, then

$$\widetilde{Z}_{r+1} \le c_{\mathrm{aff}}(a_r) Z_r + d_{\mathrm{aff}}(a_r) \le c Z_r + d. \tag{64}$$

For every iterate w_s in that terminal history and every $\chi > 0$, Young's inequality gives

$$\begin{aligned}\|w_s - w_{r+1}^\star\|^2 \le (1+\chi)\|w_s - w_r^\star\|^2 \\ + (1+\chi^{-1})\|w_{r+1}^\star - w_r^\star\|^2.\end{aligned} \tag{65}$$

Taking expectations and the maximum over the history yields

$$Z_{r+1} \le (1+\chi) c Z_r + (1+\chi) d + (1+\chi^{-1}) v^2. \tag{66}$$

Iteration proves (24) whenever $\bar{c} = (1+\chi)c < 1$.

For a one-step TD advantage,

$$\widehat{A}_w - A^\star = \{\gamma\phi(s') - \phi(s)\}^\top (w - w_n^\star). \tag{67}$$

The feature bound implies $\|\gamma\phi(s') - \phi(s)\| \le (1+\gamma)\Phi$. Cauchy–Schwarz followed by the definition of Z_n proves (25).

C. *Existence and complexity of the mixed controller*

For fixed (q, b), a_δ, β_δ, h_τ, and g_τ are continuous functions of η. The optimized Young coefficient and hence c_{aff} and d_{aff} are continuous wherever their denominators are positive, with the zero-delay values defined by continuity. The score in (26) is therefore continuous on each certified component. Restricting to its closed feasible portion gives existence by the extreme-value theorem.

The controller evaluates at most $N_q N_b$ scalar problems. If a deterministic scalar routine uses L_η score evaluations per pair, the control arithmetic is $O(N_q N_b L_\eta)$. For rank-one $H_{i,k}$, matrix–vector products can be accumulated from feature inner products without forming a dense $d \times d$ matrix. The calibration update is consequently $O(qd)$ in arithmetic and $O(d)$ in working memory, which proves Proposition 2.

Appendix C
Correlation-Limited Minimax Risk

For one round of the Gaussian location subclass, the agent average has variance $\sigma_c^2 + \sigma_e^2/q$. The T round averages are independent and Gaussian with known variance, so their sample mean is minimax for squared loss over the unrestricted location family and has risk $T^{-1}(\sigma_c^2 + \sigma_e^2/q)$. Dividing the single-agent risk by this value gives (12). Since $1+(q-1)\rho \ge q\rho$ for $\rho > 0$, the speedup is at most ρ^{-1}.

Appendix D
Proof of the Stopped Adaptive Information Identity

Condition on the pre-action history and realized action A_k. An orthogonal rotation maps X_k to the normalized mean $Y_k = q_k^{-1/2}\mathbf{1}^\top X_k$ and $q_k - 1$ contrasts. The contrasts are independent standard normal variables under both hypotheses and therefore contribute zero to the likelihood ratio.

The Kalman predictive law of Y_k under hypothesis j is $\mathcal{N}(\sqrt{q_k}\, m_{j,k}^-, V_{j,k})$. Factoring the stopped path law into action kernels and conditional observation kernels makes the common predictable action factors cancel pointwise. The log likelihood ratio is therefore the sum of scalar innovation log likelihood ratios. Conditional expectation under hypothesis zero gives the Gaussian divergence in (30). Tonelli's theorem applies because each conditional divergence is nonnegative and the budget bounds τ. This proves (34); exchanging the hypotheses proves the reverse identity.

For a decision $\widehat{J}$, data processing gives

$$\mathrm{KL}(P_j^\pi|\mathcal{F}_\tau \,\|\, P_{1-j}^\pi|\mathcal{F}_\tau) \geq \mathrm{kl}(\mathbb{P}_j\{\widehat{J}=j\}, \mathbb{P}_{1-j}\{\widehat{J}=j\}). \tag{68}$$

If both directional errors are at most δ, monotonicity of binary relative entropy yields (35).

Appendix E
Occupation-Measure Opportunity Lower Bound

Let $\Delta_j(z,a) \geq 0$ be the one-step opportunity loss relative to the instance-j certified oracle. For a stopped predictable policy, define

$$\nu_j(G,a) = \mathbb{E}_j^\pi \sum_{k \leq \tau} \mathbf{1}\{Z_k \in G, A_k = a\}. \tag{69}$$

Every δ-correct policy induces an occupation measure satisfying the controlled Kalman flow constraints,

$$\begin{aligned} \int \mathcal{I}_{j,1-j}(z,a)\nu_j(dz,da) &\geq \mathrm{kl}(1-\delta,\delta), \\ \int (h+q(a))\nu_j(dz,da) &\leq B_{\mathrm{msg}}, \\ \int b(a)\nu_j(dz,da) + D\mathbb{P}_j(\tau>0) &\leq B_{\mathrm{env}}. \end{aligned} \tag{70}$$

Hence its expected identification opportunity cost is at least the infimum of $\int \Delta_j(z,a)\nu_j(dz,da)$ over this feasible occupation set. This formulation retains the belief-state dependence created by irregular Markov gaps and changing participation.

For completeness, let $T_a(z,dz')$ be the belief-state transition kernel created by action a and the scalar Kalman innovation. The stopped occupation measure and terminal belief law ζ_j obey, for every bounded measurable test function f,

$$\int f(z)\zeta_j(dz) - f(z_1) = \int \left[\int f(z')T_a(z,dz') - f(z)\right]\nu_j(dz,da). \tag{71}$$

This is the controlled flow constraint included in $\mathcal{M}_j$. Predictability ensures that T_a is chosen before the current innovation is observed.

The expected accumulated opportunity loss is exactly

$$\mathbb{E}_j^\pi \sum_{k\leq\tau} \Delta_j(Z_k,A_k) = \int \Delta_j(z,a)\nu_j(dz,da). \tag{72}$$

Equations (34), (7), and (71) show that the occupation measure of every admissible policy belongs to $\mathcal{M}_j$. Taking an infimum over the larger set of all feasible measures gives (37) and proves Proposition 4.

Appendix F
Proof of the Main Learning-Value Guarantee

Condition on whether the likelihood test identifies instance j correctly. On the correct event, the post-probe risk is at most $\overline{R}_j^\star(B_d^D, D)$. On the wrong event, its excess over the correct catalogue optimum is at most $W_j(d)$. Since the wrong-event probability is at most δ, total expectation proves (42).

Subtracting the full-budget all-agent risk and using (39) gives

$$\mathbb{E}_1\|e_{\mathrm{LA}}\|^2 - \overline{R}_1^{\mathrm{all}}(B,D) \leq C_1^{\mathrm{probe}} + C_1^D - G_1 + \delta W_1 < 0. \tag{73}$$

The same decomposition on instance zero is bounded above by $\epsilon_{\mathrm{safe}}(d) \leq \bar{\epsilon}$. The fallback statement follows directly from the strict admission rule.

Appendix G
Proof of the Threshold Sandwich

Write $d_\delta = \mathrm{kl}(1-\delta,\delta)$. Compact separation, $\lambda \leq 1-\gamma$, and continuity of the scalar Kalman recursion imply that the directional information of every feasible observation is at most a finite class constant i_+. Every action has a positive resource cost, so the information constraint (35) implies $B_N \geq c_N d_\delta$ for a constant $c_N > 0$ depending only on the class and the budget ray.

For the fixed probe in the theorem assumptions, compactness gives a positive lower Bhattacharyya information i_- per retained observation. Thus a block of

$$n_\delta = \lceil i_-^{-1}\log(1/\delta)\rceil \tag{74}$$

has both directional errors at most δ. Its two resource costs are at most $c_p n_\delta + D_{\max}$ in budget-ray units. For $0<\delta\leq\delta_0<1/2$, continuity of $\log(1/\delta)/d_\delta$ on $(0,\delta_0]$, together with its finite limit at zero, yields

$$n_\delta \leq c_0 + c_1 d_\delta \leq c_0 + c_2 B_N. \tag{75}$$

For a stationary action $a=(q,b,\eta)$, the exact running-mean risk has the expansion

$$R_j^{\mathrm{mean}}(a;s) = \frac{K_j(a)}{s} + O_{\mathcal{K}}(s^{-2}) \tag{76}$$

uniformly along the budget ray. The same expansion is assumed for every certified catalogue risk. Removing n_δ probe blocks and at most $D_{\max}$ terminal transitions perturbs each certificate by at most

$$\frac{L_{\mathcal{K}}(n_\delta + D_{\max} + 1)}{s^2} \tag{77}$$

for a class constant $L_{\mathcal{K}}$ once all catalogue actions are feasible. The high-instance all-agent-to-oracle improvement is at least $g_{\min}/s - O_{\mathcal{K}}(s^{-2})$. By the wrong-commit assumption, the term δW_1 consumes at most half of this leading gain. Equations (75)–(77) therefore make the strict high-instance admission inequality hold when

$$s \geq c_3 B_N + c_4(B_{\mathrm{oracle}} + 1). \tag{78}$$

All low-instance certificate differences are $O_{\mathcal{K}}(s^{-1})$. For a fixed admissible safety target, they fall below that target once $s \geq B_\epsilon$. Taking the maximum of these sufficient scales and increasing class constants gives

$$B_S \leq C_{\mathcal{K}} B_N + C'_{\mathcal{K}}(B_{\mathrm{oracle}} + B_\epsilon + 1). \tag{79}$$

The inequality $B_N \leq B_S$ follows from the necessity of the two directional information constraints for every admitted δ-correct probe. This proves (43).

Appendix H
Secondary SDDE Interpretation

The controlled discrete recursion has the local stochastic delay differential equation representation

$$dE(t) = -\frac{1}{q}\sum_{i=1}^{q} A_i E(t-\tau_i(t))dt + \sqrt{\eta}\,\Sigma_q^{1/2}(E_t)dW(t). \tag{80}$$

The diffusion must retain the state-dependent quadratic variation induced by random temporal-difference Jacobians. Near the fixed point, its multiplicative component obeys a quadratic bound proportional to $K_q\|E_t\|^2$. This representation explains why correlation changes both the noise floor and the mean-square stability margin. The paper's finite-time guarantees are provided by the exact discrete-time Lyapunov analysis above.

Appendix I
Experimental Estimands and Confidence Construction

Let $0 = u_0 < u_1 < \cdots < u_L = 1$ be the registered charged-resource fractions. For seed s and policy p, $E^x_{s,p}(u_\ell)$ is the squared parameter error or initial-state return-estimation error divided by its value at u_0. The trajectory AUC is the trapezoidal integral

$$\mathrm{AUC}^x_{s,p} = \sum_{\ell=1}^{L} \frac{u_\ell - u_{\ell-1}}{2}\left[E^x_{s,p}(u_\ell) + E^x_{s,p}(u_{\ell-1})\right], \tag{81}$$

where $x \in \{\mathrm{par}, \mathrm{ret}\}$. The resource axis charges skipped transitions, actor messages, and terminal delay.

Each bootstrap replicate resamples paired seeds and recomputes the ratio of geometric-mean risks over every registered cell. The reported interval uses the 2.5–97.5 percentiles of 2,000 replicates from the frozen bootstrap seed. Pairing retains common-random-number variance reduction without treating checkpoints as independent.

Finite-value and cumulative budget checks precede aggregation. The fixed-q baseline is selected once on the disjoint development population and frozen for confirmation.

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